

Words That Move Patients

How to talk about an unproven oxygen carrier without inflating hope, hiding risk or losing the people you are trying to help

A foundation that campaigns for a medical technology holds a strange kind of power. It does not make the product, it does not regulate it, and it does not prescribe it. What it does is shape the sentences that patients, families, clinicians and legislators repeat to one another. Those sentences travel further than any trial report. They decide who becomes curious, who becomes frightened, and who decides the topic is not worth a phone call.

This essay is about the craft of those sentences. It is aimed at anyone writing about BHOC for a general audience: advocates, volunteers, journalists and the patients who post about it. It is practical, and it makes an argument that some readers will find uncomfortable: that the most effective way to advance a promising but unproven technology is to be visibly, boringly, consistently honest about what is not yet known.

Why the words matter more here than usual

Three features of the BHOC field raise the stakes of language.

First, the history is bad. Products in this class have been associated with harm in trials, and a pooled analysis in 2008 combined sixteen trials of several products and found a higher risk of death and of heart attack in treated patients. Any reader who has heard about that history arrives with suspicion. Careless enthusiasm confirms the suspicion.

Second, the audience is vulnerable. People who search for an oxygen carrier are often people or families in trouble: someone who cannot receive blood, someone in a long illness, someone frightened. They are inclined to hear hope in any sentence that offers it. That inclination puts a moral duty on the writer.

Third, the science is subtle. Terms like oxygen affinity, cell-free hemoglobin and vasoconstriction have precise meanings, and their casual use creates false impressions. A word like artificial blood, for instance, suggests a complete replacement for something that a bovine hemoglobin solution is not.

The three-tier rule

The simplest tool I know for honest communication is to sort every claim into one of three tiers and to mark the tier in the sentence.

Tier one is what is established. Things that have been measured, in humans, reproducibly, and reported in the peer-reviewed literature. For example: a bovine hemoglobin carrier has been approved for anemic dogs, and has been approved for human surgical anemia in South Africa. Those are facts with sources.

Tier two is what is plausible. Claims that follow from mechanism or from animal data or from small studies, and that have not been confirmed in humans. For example: a cell-free carrier

might reach tissue that red cells cannot when a vessel is narrowed. It is a reasonable hypothesis, supported by physiology and by experiments, and it is not proven as a clinical benefit.

Tier three is what is hoped. Aspirations about what the technology might one day do. For example: a shelf-stable carrier might someday allow emergency oxygen delivery in remote places. That is a goal, and it should be written as one.

The rule is that no sentence may present a tier-two or tier-three claim in the grammar of a tier-one claim. The words that carry the tier are small: is, may, might, has been shown, is hoped. Their precision is the whole point. A page that swaps them casually, turning may into does, has crossed from advocacy into promotion.

Numbers that mislead and numbers that inform

Patients are asked to reason about risk, and the way numbers are presented can push their reasoning in either direction.

Take the 2008 pooled analysis. It is often summarized as showing a thirty percent increase in the risk of death and nearly a threefold increase in the risk of heart attack. Those are relative risks. A relative risk is a ratio, and a ratio alone says nothing about how many people are affected. If the baseline risk of an event is one in a thousand, tripling it gives three in a thousand: a large relative increase and a small absolute one. If the baseline is one in ten, tripling it gives three in ten, which is a catastrophe.

The honest presentation gives both numbers: the relative change and the absolute counts, in the same population, over the same time. Statisticians who study risk communication have found that people understand natural frequencies, such as three out of a thousand, much better than percentages or probabilities. A sentence like out of every thousand patients treated, this many more had a heart attack is clearer than a hazard ratio.

The same care applies to favorable numbers. A company may report that most patients receiving a carrier survived. Most compared with what? A group of patients who could not receive blood, treated under expanded access, will have a survival rate that reflects the severity of their condition and the selection of who was chosen. Without a comparison group, survival numbers are not evidence of benefit. They are descriptions.

I would propose a plain rule: any number about outcomes should be accompanied by its denominator, its comparison, and its source, in the same sentence or the next one. If any of the three is unavailable, say so.

The vocabulary problem

Some words are especially treacherous.

Artificial blood. The product is not blood, and it does not do what blood does. It carries oxygen, and little else. It does not clot, it does not fight infection, and it does not last. Blood substitute has the same problem. Oxygen carrier or oxygen therapeutic is more accurate.

Safe. A word to avoid. Safe compared with what, at what dose, in whom, and for how long? A product can be reasonably safe for a defined use and unsafe for another. Say what was observed.

Approved. As described in another post, this needs a regulator, a country and an indication.

Miracle, breakthrough, revolutionary. Words that belong on a marketing page, not a fact sheet. They cannot be verified, and they attract exactly the reader you are trying not to mislead.

Cure. A carrier treats a condition, at best. It does not cure anemia or a disease of the vessels, and calling it a cure is untrue.

Natural. Bovine hemoglobin comes from an animal. It is purified and chemically modified. The word invites assumptions about safety that the science does not support, and it is irrelevant to whether the product works.

Not a word but a habit: the passive voice that hides who found what. It has been shown that is weaker than a 2021 randomized trial in the New England Journal of Medicine found that. The reader deserves to know who said it.

Conflicts of interest, said early

An advocacy organization that works with a developer, receives support from a company or shares goals with one, has an obligation to say so at the start of every relevant communication, not at the bottom of a page.

The obligation is not about wrongdoing. Collaboration is normal and often essential. It is about giving readers what they need to weigh what they are told. A sentence like this organization is working with the developer on outreach and has not received payment for its statements, or has received support, is short and settles a question that would otherwise linger.

Readers are more forgiving of a disclosed relationship than of a discovered one. In fields with a history of financial misstatement, as this one has, credibility is a scarce resource and disclosure is how it is earned.

Writing for the frightened reader

Some of your readers will be people in crisis. They need a different register from an analyst.

Lead with what they can do. If you are describing a route to access, put the practical steps first: whom to ask, what documents are needed, what the physician must do. Then give the context.

Be gentle about prognosis. You do not know their case. Avoid statements that predict, and refer to the treating team for anything specific.

Avoid pressure. Never suggest that a patient should demand a specific product. The decision belongs to the patient and the clinician together, and the person on the other end of the page may have a doctor who knows things you do not.

Say plainly that the product is not available everywhere, may not be appropriate, and carries risks. A short paragraph of this kind, placed high on the page, is not a legal disclaimer. It is respect.

Writing for the skeptical reader

Others will be doctors, scientists, reporters and officials who have seen bright promises fail. They read looking for the overreach.

Give them sources. A claim followed by a reference can be checked. A claim without one is a tell.

Give them the bad news yourself. Name the 2008 analysis, the trials that failed, the companies that collapsed. Do it before they do. A page that acknowledges the record earns the right to be heard on the rest.

Give them the gaps. State what is not known. An honest description of the missing evidence is more persuasive to a skeptic than a stack of encouraging anecdotes.

And leave out the adjectives. A skeptical reader treats adjectives as noise.

A model fact sheet

To make this concrete, here is the structure I would use for a one-page fact sheet, described in words so that it can be adapted.

Begin with a single sentence that says what the product is and what it is not. Follow with a short paragraph on where it stands: which regulators have acted, in which countries, for what use, and its status in the country of the reader. Then a paragraph on what the evidence shows, tier by tier, with sources. A paragraph on the known risks and unknowns. A paragraph on how a patient might access it, if at all, and what to ask a clinician. A line of disclosure about the relationship of the publisher to any developer. And a date, because the facts change.

Keep it to a page. If it is longer, most readers will not finish it, and the ones who do not finish it will remember the first paragraph.

One result, told three ways

An example makes the difference tangible. Imagine a small study of a carrier given to twenty patients who could not receive blood, of whom fifteen survived to hospital discharge. This is invented for illustration and does not describe any real study.

The promotional telling: Seventy-five percent of patients survived thanks to the carrier, a breakthrough for people who cannot receive blood. Every clause is either unsupported or misleading. Thanks to implies a cause the study cannot show. Breakthrough is a verdict, not a finding.

The cautious but empty telling: A small study reported some survivors and some deaths, and more research is needed. Nothing false, and nothing a reader can use.

The honest telling: In a series of twenty patients who could not receive blood and who were given the carrier under expanded access, fifteen were alive at discharge and five died. The study

had no comparison group, so it cannot show whether the carrier changed the outcome. Patients differed widely in the cause and severity of their anemia. The authors reported the following adverse events. The series suggests the product can be given in this setting; it does not show that it saves lives.

The third version is longer, and every sentence in it is one a journalist, a clinician or a regulator can check. It is also the version that survives a hostile question. Writing it takes discipline, but the discipline is mechanical: state the number and the denominator, state what is missing, state what the result does and does not show.

When you do not know the answer

Advocates get asked questions they cannot answer. Is it safe for my father? Will insurance cover it? Should I ask the surgeon for it tomorrow?

The temptation is to say something reassuring. The better response has three parts. Say what you do know, briefly. Say what you do not know and cannot know from where you sit. And say who can help: the treating physician, the hospital ethics service, the manufacturer's medical information line for an expanded access request.

There is nothing shameful in the second part. People who ask hard questions in hard moments usually know that no one can give them certainty. What they resent is being handled. A plain statement of the limits of your knowledge is a form of respect, and it is remembered.

It helps to prepare a short list of the questions you cannot answer and the referrals for each, and to keep it where every volunteer can see it. Consistency matters: two volunteers giving different answers to the same question is worse than either giving a modest one.

The short-form problem

Social media compresses. A post of two hundred and eighty characters or a short video cannot hold a three-tier evidence framework. Yet it is where many readers first meet the topic.

I would suggest a few rules. Do not make a claim in a post that you would not make in a fact sheet. Link the sentence to the sheet, and say in the post itself that the link has more detail. Prefer questions to declarations: What would it take for a cell-free oxygen carrier to be tested in people who cannot receive blood? invites readers in without promising anything. Avoid superlatives and avoid the word safe. And never share a testimonial as evidence. A patient's story can be told with compassion, but it should be labeled as a story, not a result.

When a post turns out to be wrong, correct it visibly and quickly. A short reply that says we got this wrong, here is the correct figure, and here is the source, costs a little pride and buys a lot of trust. Deleting quietly costs the trust and saves the pride, and it usually leaks.

The correction habit

Every organization that communicates about science makes errors. The measure of quality is not the absence of errors but what happens after. I recommend a few habits.

Keep a public corrections note. Date each entry, state what was wrong and what is right.

Date every page. A statement that was accurate a year ago may not be now, and readers deserve to know how old a claim is.

Assign someone to check citations. A second pair of eyes catches transposed numbers and misremembered years, the two most common errors in this kind of writing, and I have made both in the course of preparing these essays.

Invite outside review. A clinician, a statistician and a patient reading your draft before it goes out will each catch different things.

A word on tone

A last point, not about accuracy. The tone of a page carries information. Anger reads as certainty; a steady, curious voice reads as reliability. Patients have good reasons to be angry about the state of care, and advocacy is often born from that anger. It is a legitimate fuel. But when the goal is to convince a regulator or a clinician, the anger is best kept in the motivation and out of the prose.

This does not mean blandness. Vivid, specific writing, a real patient's story told with consent, a well-chosen number, a clear picture of a mechanism, is the opposite of bland. It means that force should come from precision, not volume.

What good communication buys

I have been arguing for restraint, and I should say what restraint gets you.

It gets credibility, which is the only thing that lets an advocacy organization be invited into rooms where decisions are made. Regulators, clinicians and legislators talk to groups that they trust not to embarrass them.

It gets better participation in trials. Patients who understand what a trial can and cannot offer enroll with realistic expectations, drop out less and report more accurately.

It gets protection. When something goes wrong in a field, as in any field, the organizations that overpromised are the ones that face the anger. The ones that were honest can say we told you this was uncertain, and mean it.

And it gets a longer life for the whole enterprise. The history of oxygen carriers is partly a history of enthusiasm outrunning evidence, followed by disillusionment and withdrawal of support. A field where advocates keep their claims small and their sources visible is a field less likely to boom and bust.

A final practical suggestion is to test your own materials on a reader who has no stake in the cause. Hand a fact sheet to a friend who is a nurse, a teacher or a mechanic, ask them to explain it back in their own words, and listen for the places where their version drifts from yours. Those are the sentences that need rewriting, and no amount of expertise substitutes for the discovery.

Closing

The paradox of advocacy is that the strongest statement is often the most modest. A sentence that says here is what we know, here is what we do not, and here is what we are doing to find out is not a weak sentence. It is the one that can survive contact with a skeptical reader on a bad day.

Words move patients. They can move them toward a decision they will regret or toward one they will understand. That responsibility does not go away because the cause is good. It grows.

Further reading

Guidance on communicating risk with natural frequencies and on presenting absolute and relative changes together is available from the risk communication literature, including work by Gerd Gigerenzer and colleagues. The 2008 pooled analysis of hemoglobin carrier trials was published in the Journal of the American Medical Association. Readers preparing public materials should also consult their regulators' guidance on communications about investigational products, which restrict claims of safety and effectiveness.